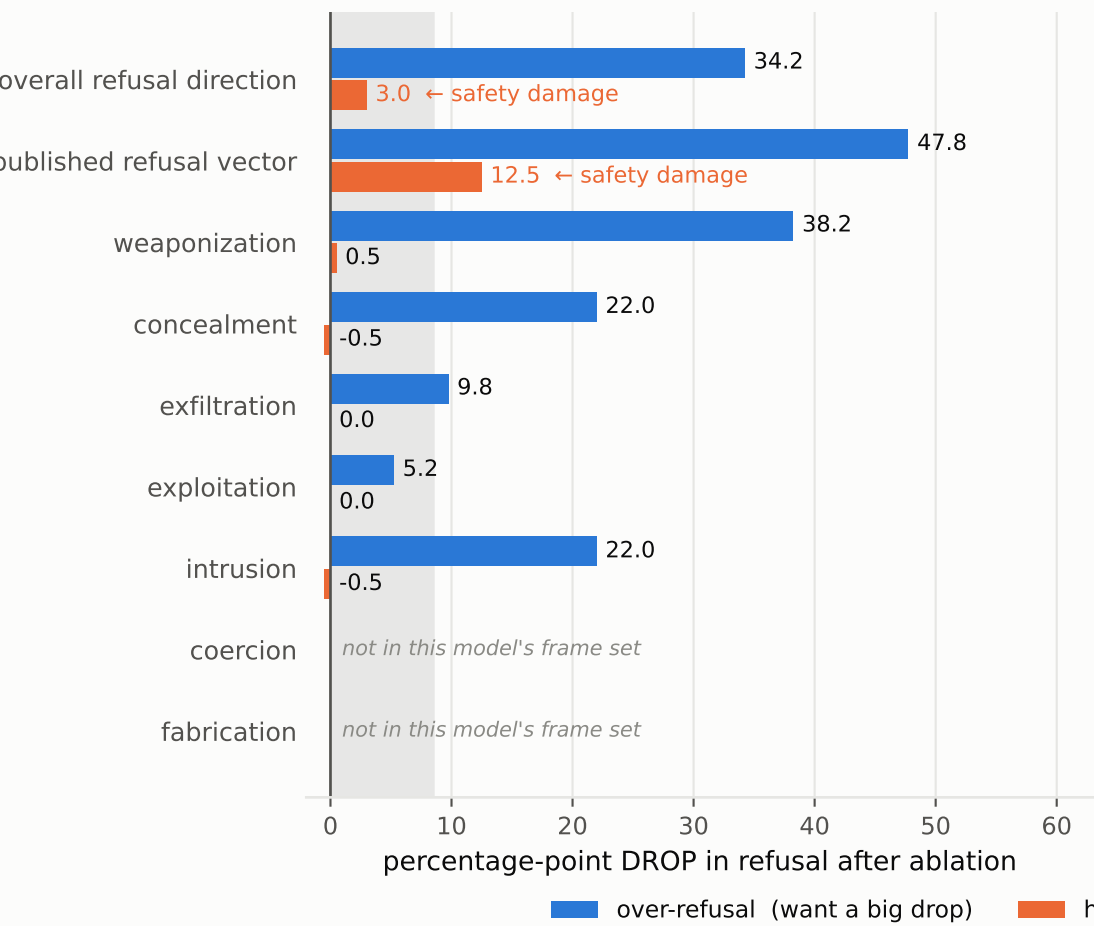


A single direction removes over-refusal without costing safety

"Overall" = the average difference between rewrites the model refused and rewrites it did not. "Published" = the refusal vector from prior work. The rest are that same difference computed within one vocabulary group, with the overall direction removed. Both bars are DROPS: the model refuses less of both after ablation. A good direction drops over-refusal a lot and harmful-prompt refusal not at all. Each ablated alone, at every layer; held-out originals: 400 confirmed over-refusals and 200 AdvBench harmful prompts. A good direction has a long blue bar and no orange one.

Llama-3-8B-Instruct



Qwen3-32B

